

Good, Better, Best (1)

Input file: standard input
Output file: standard output
Time limit: 1 second
Memory limit: 256 megabytes

This is Version (1) of the problem. The only difference between this version and Version (2) is the constraints.

An array is called **GOOD** if there exists an element x in the array such that x divides every element in the array*.

An array is called **BETTER** if it is **GOOD** and such an element x **appears exactly once** in the array.

An array is called **BEST** if it is **BETTER** and the sum of the array elements is divisible by the sum of all digits of the array elements.

Given an array a of n positive integers, determine its **highest characteristic**.

The characteristics are ordered as follows:

$$\text{GOOD} < \text{BETTER} < \text{BEST}$$

Read the note section below for better understanding.

Input

The first line contains an integer n ($1 \leq n \leq 1000$).

The second line contains n integers a_i ($1 \leq a_i \leq 10^6$).

Output

Print the highest characteristic of the array or "**BAD**".

where BAD means that the array is not **GOOD**.

Examples

standard input	standard output
1 7	BEST
3 3 6 12	BETTER
3 2 4 12	BEST
4 6 10 15 30	BAD

Note

The characteristics are ordered as follows:

$$\text{GOOD} < \text{BETTER} < \text{BEST}$$

That is, every **BEST** array is also **BETTER**, and every **BETTER** array is also **GOOD**. However, you must print only the **highest characteristic** satisfied by the array.

*A number x divides a number y if $y \bmod x = 0$.

For example:

- The array $[2, 2, 4, 8]$ is **GOOD** because 2 divides every element. It is not **BETTER** because 2 appears more than once.
- The array $[2, 4, 8]$ is **BETTER** because 2 divides every element and appears exactly once.
- The array $[2, 4, 12]$ is **BEST** because it is **BETTER**, and:
[$2+4+12 = 18$]
while the sum of all digits is:
[$2+4+1+2 = 9$]
Since $18 \bmod 9 = 0$, the array is **BEST**.